Unit S4 Statistics 4

GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional)
A2 optional unit

S4.1 Unit description

Quality of tests and estimators; one-sample procedures; two-sample procedures.

S4.2 Assessment information

Prerequisites

A knowledge of the specification for S1, S2 and S3 and its prerequisites and associated formulae is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. The paper will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+, -, \times, \div, \pi, x^2, \sqrt{x}, \frac{1}{x}, x^y, \ln x, e^x, x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

Type I error: $\mathrm{P}(\text{reject } \mathrm{H}_0 \mid \mathrm{H}_0 \text{ true})$

Type II error: $\mathrm{P}(\text{do not reject } \mathrm{H}_0 \mid \mathrm{H}_0 \text{ false})$

1 Quality of tests and estimators

What students need to learn:

Type I and Type II errors. Size and Power of Test. The power test.	Simple applications. Calculation of the probability of a Type I or Type II error. Use of Type I and Type II errors and power function to indicate effectiveness of statistical tests. Questions will not be restricted to the Normal distribution.
Assessment of the quality of estimators.	Use of bias and the variance of an estimator to assess its quality. Consistent estimators.

---

2 One-sample procedures

What students need to learn:

Hypothesis test and confidence interval for the mean of a Normal distribution with unknown variance.	Use of $\frac{\bar{X}-\mu}{S/\sqrt{n}} \sim t_{n-1}$. Use of t-tables.
Hypothesis test and confidence interval for the variance of a Normal distribution.	Use of $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$. Use of χ^2 tables.

3 Two-sample procedures

What students need to learn:

Hypothesis test that two independent random samples are from Normal populations with equal variances.	Use of $\frac{S_1^{\,2}}{S_2^{\,2}} \sim \mathrm{F}_{n_1-1,\, n_2-1}$ under H_0. Use of the tables of the F-distribution.
Use of the pooled estimate of variance.	$s^2 = \frac{(n_1-1)s_1^{\,2} + (n_2-1)s_2^{\,2}}{n_1+n_2-2}$
Hypothesis test and confidence interval for the difference between two means from independent Normal distributions when the variances are equal but unknown.	Use of t-distribution. $\frac{\bar{X}-\bar{Y}}{S\sqrt{\frac{1}{n_x}+\frac{1}{n_y}}} \sim t_{n_x+n_y-2}$, under H_0.
Paired t-test.	